Alan George

Curriculum Vitae

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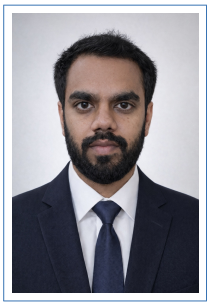
Date of Birth: 16.12.1995
Nationality: Indian *Marital Status: Unmarried*



Professional Summary

Ph.D. researcher in Computational Materials Science with expertise in molecular dynamics simulations, polymer–drug systems, and machine learning for solubility prediction. Published first-author and co-author research in Wiley, Elsevier, and RSC journals. Experienced in Python, SQL, FastAPI, HPC workflows, and deployment of scientific machine learning tools for computational chemistry and materials informatics applications.

Education

2024–Present **Ph.D. Computational Material Science**, *Friedrich Schiller University*, Germany.

2017–2019 **M.Sc. in Physics**, *NIT Jalandhar*, India.

2014–2017 **B.Sc. (Hons) Physics**, *St. Stephen's College, University of Delhi*, India.

2012–2014 **Senior Secondary (CBSE)**, *Delhi*, India.

Research Experience

2024–Present **Ph.D. Researcher**, *Friedrich Schiller University*, Jena, Germany.

- Performed molecular dynamics simulations on polymers, small molecules, and polymer–drug systems to study solubility and formulation behavior.
- Developed machine learning models for predicting Hildebrand solubility parameters using molecular dynamics simulation data.
- Screened polymer carriers including PLGA, polyesteramides, and acetylated dextran for anti-inflammatory compounds such as 6BIGOE, MF-15, and BRP-201, supporting formulation selection and validation with experimental collaborators.
- Developed and deployed machine learning tools on the **PolyDrug** web platform for Hildebrand solubility prediction, Gibbs free energy visualization, and prediction of Flory–Huggins interaction parameters from SMILES representations.

Publications

- **George, A.**, & Sierka, M. (2026). *Machine Learning Meets Molecular Dynamics: Accurate Prediction of Polymer Solubility Parameters*. *Advanced Theory and Simulations* (Wiley, 2026).
- Westhoff, J., Weber, C., Bachmann, V., Göppert, N. E., Peric, M., **George, A.**., Sierka, M., Hoeppener, S., Vilotijevic, I., Schubert, U. S., Werz, O., Fischer, D. (2025). *Amino acid-equipped Poly (ester amide) nanoparticles for improved encapsulation efficiency of lipophilic anti-inflammatory 6BIGOE based on molecular dynamic simulations*. *International Journal of Pharmaceutics* (Elsevier, 2025).
- Klepsch, L. C., Dahlke, P., Behnke, M., Tsarenko, E., Göppert, N. E., Klemm, P., Meyer, J., **George, A.**, Chi, M., Czapplewska, J., Vollrath, A., Weber, C., Jordan, P. M., Schubert, S., Hoeppener, S., Nischang, I., Sierka, M., Werz, O., Schubert, U. S. (2025). *Matching drug and polymer for efficient delivery of anti-inflammatory drugs: PLGA, polyesteramides, and acetylated dextran*. *Journal of Materials Chemistry B* (The Royal Society

of Chemistry, 2025).

International Conference Visits

- Poster Presentation at **Turbomole: Today and Tomorrow** (Oxford University, UK, 2024) — international meeting bringing together Turbomole developers and users to discuss theoretical developments, applications, and future directions.
- Poster Presentation at **WATOC 2025** (World Association of Theoretical and Computational Chemists, Oslo, Norway, 2025) — one of the world's largest international conferences in theoretical and computational chemistry.

M.Sc. Thesis

2018 – 2019 **Band Structure and Vibrational Properties of Graphene using Density Functional Theory (DFT)**, *Supervisor: Prof. Dr. Abhinav Pratap Singh*, NIT Jalandhar, Punjab, India.

Technical Skills

Programming	Python, SQL, FastAPI
Scientific Computing	Molecular Dynamics, DFT, DPD Simulations, HPC, Linux/Command line
Scientific Software	Material Studio, ORCA, Quantum Espresso, Turbomole
Data & ML	Machine learning, data analysis, scientific visualization
Languages	German (A2 certified, B1 in progress), English (IELTS B2)

References

Prof. Dr. Marek Sierka.
Professor, Department of Physics, Friedrich Schiller University Jena, Germany (Ph.D. Supervisor)
Email: marek.sierka@uni-jena.de

Prof. Dr. Abhinav Pratap Singh.
Associate Professor, Department of Physics, NIT Jalandhar, India (M.Sc. Thesis Supervisor)
Email: jaiswarsinghap@nitj.ac.in

Prof. Dr. Harleen Dahiya.
Associate Professor, Department of Physics, NIT Jalandhar, India (Head of Department, 2019)
Email: dahiyah@nitj.ac.in